MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – TEST (SET - B) (SOLUTIONS)** **Time:** 1 Hour
Class: Class X (CBSE) **Chapter 5: Arithmetic Progressions** **MM:** 30

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. Partial marks are allocated step-wise strictly according to CBSE board evaluation norms as shown in brackets.
3. Every proof highlights stated assumptions, substitutions, and concluding justification.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. For what value of p will $(3p-1)$, $(3p+5)$, and $(5p+1)$ be three consecutive terms of an AP? [1]

(A) 4 (B) 5 (C) 2 (D) 3

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 5** [0.5 Mark]

Applying the consecutive AP property:

Formula / Theorem Used: If a, b, c are in AP, then $2b = a + c$

Justification: [0.5 Mark]

$2(3p+5) = (3p-1) + (5p+1) \implies 6p+10 = 8p \implies 2p = 10 \implies p = \mathbf{5}$. (Terms: $14, 20, 26$.)

Q2. The next term of the AP $\sqrt{7}, \sqrt{28}, \sqrt{63}, \ldots$ is: [1]

(A) $\sqrt{70}$ (B) $\sqrt{84}$ (C) $\sqrt{98}$ (D) $\sqrt{112}$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(D) $\sqrt{112}$** [0.5 Mark]

Simplifying radicals: $\sqrt{7}, 2\sqrt{7}, 3\sqrt{7}, \ldots$ ($a = \sqrt{7}, d = \sqrt{7}$).

Formula / Theorem Used: $a_4 = a + 3d$

Justification: [0.5 Mark]

$a_4 = \sqrt{7} + 3\sqrt{7} = 4\sqrt{7} = \sqrt{16 \times 7} = \mathbf{\sqrt{112}}$.

Q3. In an AP, if $a_{20} - a_{15} = 20$, then its common difference d is: [1]

(A) 4 (B) 5 (C) 6 (D) 3

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 4** [0.5 Mark]

Using $a_n = a + (n-1)d$:

Formula / Theorem Used: $a_{20} - a_{15} = (a + 19d) - (a + 14d) = 5d$

Justification: [0.5 Mark]

$5d = 20 \implies d = \frac{20}{5} = \mathbf{4}$.

Q4. **Assertion (A):** The common difference of the AP $5, 1, -3, -7, \ldots$ is 4.
Reason (R): The common difference of an AP is given by $d = a_n - a_{n-1}$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: (D) Assertion (A) is false, but Reason (R) is true. [0.5 Mark]

Checking the sign of consecutive differences:

Formula / Theorem Used: $d = a_2 - a_1 = 1 - 5 = -4 \neq +4$

Justification: [0.5 Mark]

Reason (R) states the true formula $d = a_n - a_{n-1}$. Applying it gives $d = 1 - 5 = -4$, NOT 4. Thus (A) is false and (R) is true, so option **(D)** holds.

Q5. **Assertion (A):** The 8th term of the AP $\frac{1}{m}, \frac{1+2m}{m}, \frac{1+4m}{m}, \ldots$ is $\frac{1+14m}{m}$.
Reason (R): The nth term of an AP is given by $a_n = a + (n-1)d$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: (A) Both (A) and (R) are true, and (R) explains (A). [0.5 Mark]

Here $a = \frac{1}{m}$ and $d = \frac{1+2m}{m} - \frac{1}{m} = 2$:

Formula / Theorem Used: $a_8 = \frac{1}{m} + (8-1)(2) = \frac{1}{m} + 14 = \frac{1+14m}{m}$

Justification: [0.5 Mark]

Reason (R) is true and substituting $a = \frac{1}{m}, d = 2, n = 8$ directly proves Assertion (A).

Q6. Find the 11th term from the last term (towards the first term) of the AP $10, 7, 4, \ldots, -62$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying l and d [1 Mark]

$a = 10$, $d = 7 - 10 = -3$, last term $l = -62$.

Formula / Theorem Used: n^{th} term from end $= l - (n-1)d$

Step 2: Evaluating 11th Term from the Last [1 Mark]

$$11^{\text{th}} \text{ term from last} = -62 - (11-1)(-3) = -62 - 10(-3) = -62 + 30 = \boxed{\mathbf{-32}}$$

Q7. How many natural numbers between 101 and 999 are divisible by both 2 and 5? **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying AP of Multiples of LCM$(2,5) = 10$ [1 Mark]

Numbers strictly between 101 and 999 divisible by 10: $110, 120, \ldots, 990$ ($a = 110$, $d = 10$, $a_n = 990$).

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 2: Solving for n [1 Mark]

$$110 + (n-1)(10) = 990 \implies 10(n-1) = 880 \implies n - 1 = 88 \implies \boxed{\mathbf{n = 89}}$$

Q8. Check whether -150 is a term of the AP $11, 8, 5, 2, \ldots$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Solving for Term Index n [1 Mark]

$a = 11$, $d = 8 - 11 = -3$. Setting $a_n = -150$:

Formula / Theorem Used: $a_n = a + (n-1)d$ requires $n \in \mathbb{N}$

$$11 + (n-1)(-3) = -150 \implies -3(n-1) = -161 \implies n = \frac{161}{3} + 1 = \frac{164}{3} = 54\frac{2}{3}$$

Step 2: Conclusion [1 Mark]

Since $n = \frac{164}{3}$ is not a positive integer, **No, -150 is NOT a term of the AP**.

Q9. How many terms of the AP $27, 24, 21, \ldots$ should be taken so that their sum is zero? **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Forming Sum Equation $S_n = 0$ [1 Mark]

$a = 27$, $d = 24 - 27 = -3$, $S_n = 0$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 2: Solving for $n \geq 1$ [1 Mark]

$$\frac{n}{2}[2(27) + (n-1)(-3)] = 0 \implies n(57 - 3n) = 0 \implies 3n = 57 \quad (n \neq 0) \implies \boxed{\mathbf{n = 19} \text{ terms}}$$

Q10. If the sum of the first n terms of an AP is $S_n = 3n^2 + 4n$, find its 7th term (a_7). **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $S_n = 3n^2 + 4n$.

Formula / Theorem Used: $a_7 = S_7 - S_6$

Step 1: Computing S_7 and S_6 [1 Mark]

$S_7 = 3(7)^2 + 4(7) = 3(49) + 28 = 147 + 28 = 175$.

$S_6 = 3(6)^2 + 4(6) = 3(36) + 24 = 108 + 24 = 132$.

Step 2: Finding a_7 [1 Mark]

$$a_7 = 175 - 132 = \boxed{\mathbf{43}}$$

Q11. The sum of four consecutive numbers in AP is 32 and the ratio of the product of the first and last terms to the product of the two middle terms is 7 : 15. Find the numbers. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Choosing Symmetric Terms and Finding a [1.5 Marks]

Formula / Theorem Used: Four consecutive terms in AP: $(a-3d),\ (a-d),\ (a+d),\ (a+3d)$

$(a-3d)+(a-d)+(a+d)+(a+3d)=32 \implies 4a=32 \implies a=8.$

Step 2: Solving Product Ratio Equation for d [2.5 Marks]

$$\frac{(8-3d)(8+3d)}{(8-d)(8+d)}=\frac{7}{15} \implies \frac{64-9d^2}{64-d^2}=\frac{7}{15} \implies 15(64-9d^2)=7(64-d^2)$$

$$960-135d^2=448-7d^2 \implies 512=128d^2 \implies d^2=4 \implies d=\pm 2$$

Step 3: Listing the Four Numbers [1 Mark]

With $d=2$: $2, 6, 10, 14$ (with $d=-2$: $14, 10, 6, 2$).

$$\boxed{\textbf{2},\ \textbf{6},\ \textbf{10},\ \textbf{14}\ \text{(or }\textbf{14},\ \textbf{10},\ \textbf{6},\ \textbf{2}\text{)}}$$

Q12. How many terms of the AP $9, 17, 25, \ldots$ must be taken to give a sum of 636? **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Forming Quadratic Equation in n [1.5 Marks]

Here $a=9$, $d=17-9=8$, $S_n=636$.

Formula / Theorem Used: $S_n=\frac{n}{2}\left[2a+(n-1)d\right]$

$$\frac{n}{2}[2(9)+(n-1)(8)]=636 \implies n[9+4(n-1)]=636 \implies n(4n+5)=636 \implies 4n^2+5n-636=0$$

Step 2: Solving by Middle-Term Splitting [2.5 Marks]

Since $4\times(-636)=-2544=53\times(-48)$ and $53-48=5$:

$$4n^2-48n+53n-636=0 \implies 4n(n-12)+53(n-12)=0 \implies (n-12)(4n+53)=0$$

Step 3: Selecting Valid Natural Root [1 Mark]

Since number of terms $n\in\mathbb{N}$, reject $n=-\frac{53}{4}$, leaving:

$$\boxed{\textbf{n = 12}\ \text{terms}}$$

Q13. A manufacturer of TV sets produced 600 sets in the 3rd year and 700 sets in the 7th year. Assuming that the production increases uniformly by a fixed number every year, find: (i) the production in the 1st year, **[1.5]**
(ii) the production in the 10th year, and **[1.5]**
(iii) the total production in the first 7 years. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Annual production forms an AP with first term a and uniform annual increase d. Given $a_3 = 600$ and $a_7 = 700$.

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + a_n)$

(i) Production in the 1st Year (a) [1.5 Marks]

$a + 2d = 600$ — (1) and $a + 6d = 700$ — (2). Subtracting (1) from (2):

$$4d = 100 \implies d = 25 \implies a + 2(25) = 600 \implies a = 600 - 50 = \mathbf{550} \text{ TV sets}$$

(ii) Production in the 10th Year (a_{10}) [1.5 Marks]

$$a_{10} = 550 + (10-1)(25) = 550 + 225 = \mathbf{775} \text{ TV sets}$$

(iii) Total Production in First 7 Years (S_7) [2 Marks]

$$S_7 = \frac{7}{2}(a_1 + a_7) = \frac{7}{2}(550 + 700) = \frac{7}{2} \times 1{,}250 = 7 \times 625 = \mathbf{4{,}375} \text{ TV sets}$$

(i) **550** sets, (ii) **775** sets, (iii) **4,375** sets

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